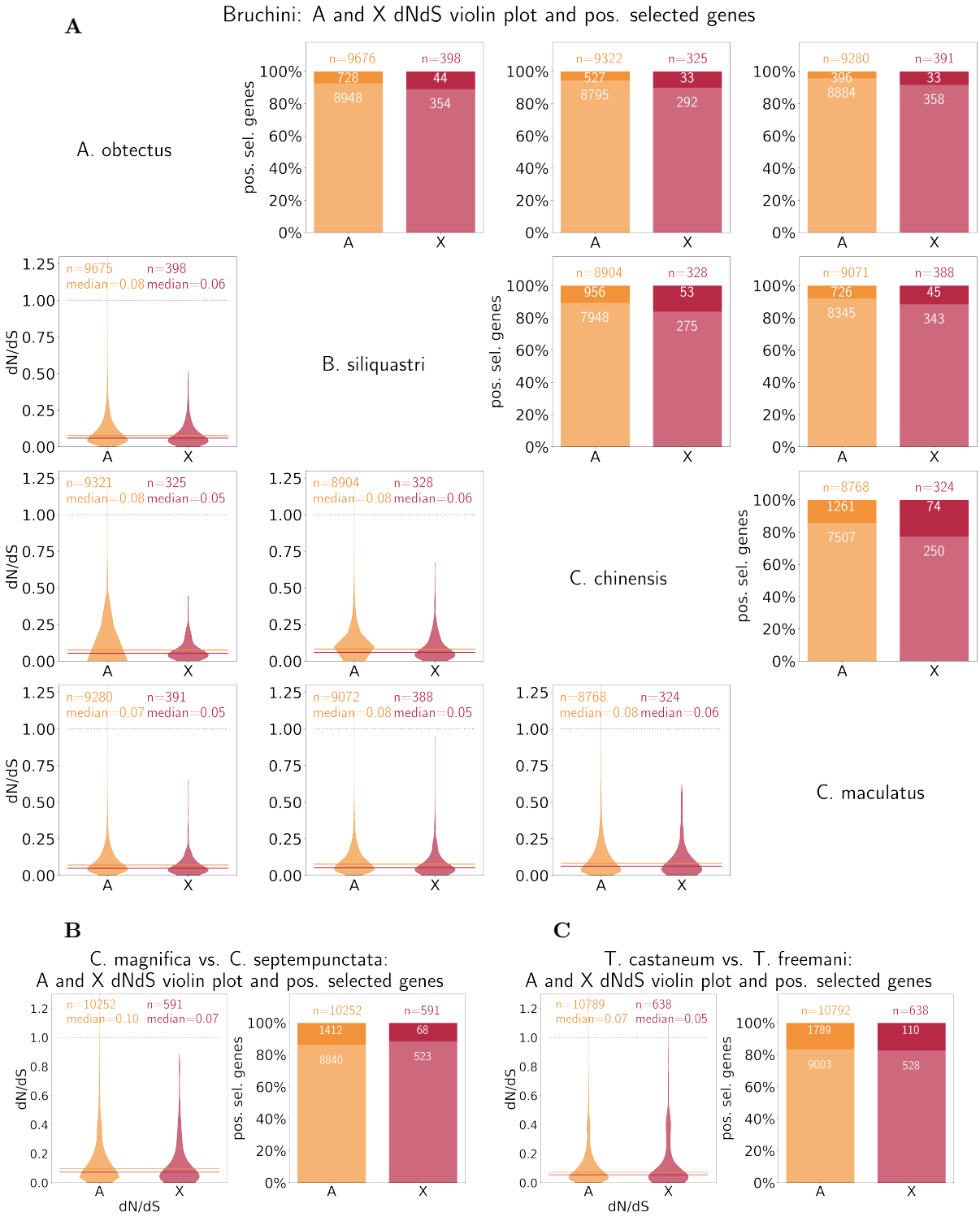
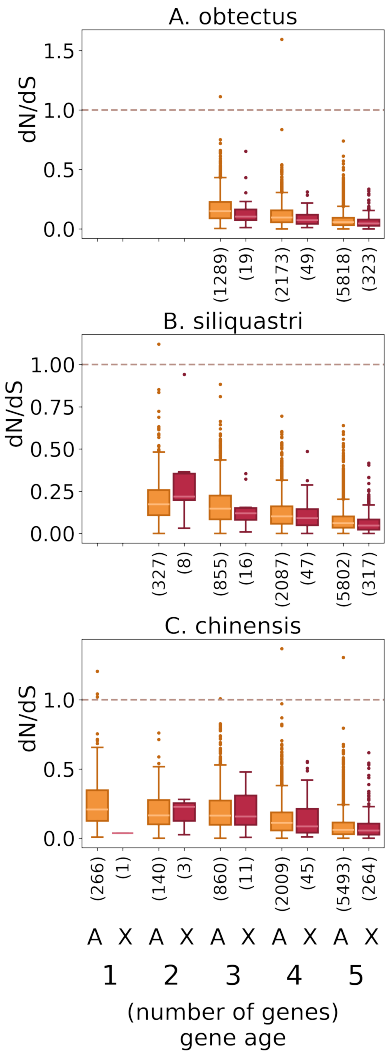
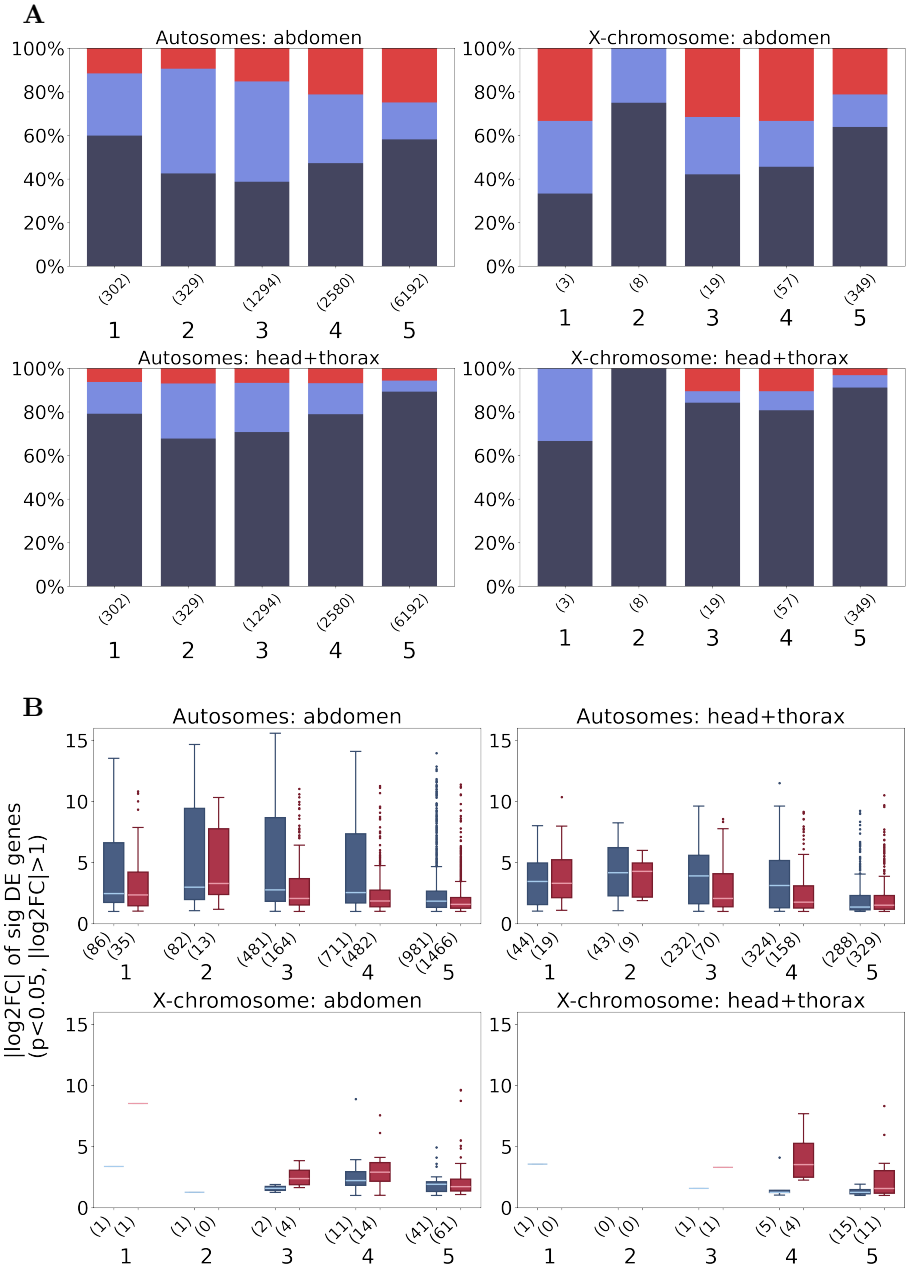
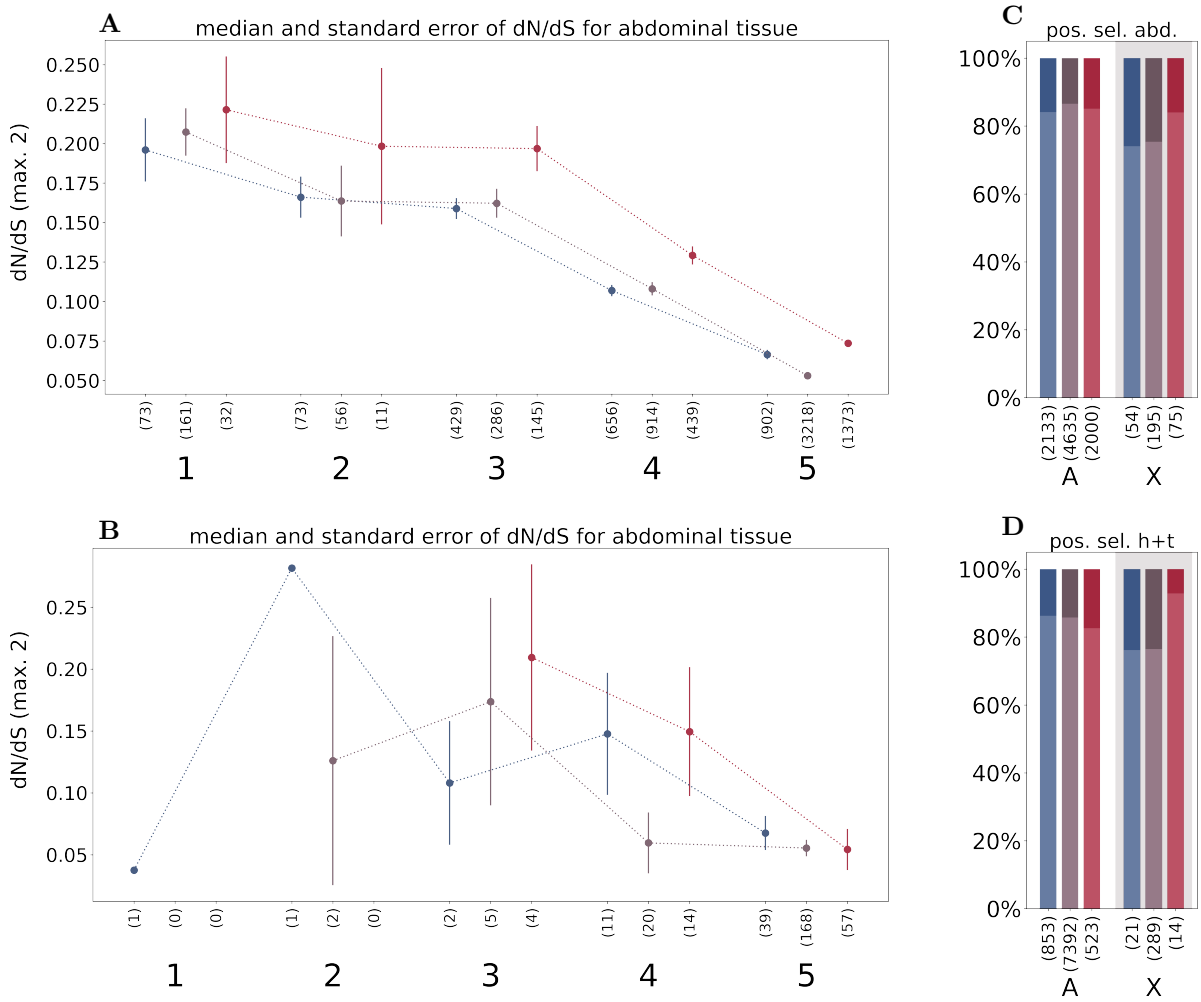

A
Bruchini: A and X dNdS violin plot and pos. selected genes
A. obtectus
n=9676 n=398
728 44
8948 354
n=9322 n=325
527 33
8795 292
n=9280 n=391
396 33
8884 358
B. siliquastri
n=9675 n=398
median=0.08 median=0.06
n=8904 n=328
956 53
7948 275
n=9071 n=388
726 45
8345 343
C. chinensis
n=9321 n=325
median=0.08 median=0.05
n=8904 n=328
median=0.08 median=0.06
n=8768 n=324
1261 74
7507 250
C. maculatus
n=9280 n=391
median=0.07 median=0.05
n=9072 n=388
median=0.08 median=0.05
n=8768 n=324
median=0.08 median=0.06
dN/dS
pos. sel. genes
A
X
B
C. magnifica vs. C. septempunctata:
A and X dNdS violin plot and pos. selected genes
n=10252 n=591
median=0.10 median=0.07
n=10252 n=591
1412 68
8840 523
C
T. castaneum vs. T. freemani:
A and X dNdS violin plot and pos. selected genes
n=10789 n=638
median=0.07 median=0.05
n=10792 n=638
1789 110
9003 528


A. obtectus
dN/dS
1.5
1.0
0.5
0.0
(1289)
(19)
(2173)
(49)
(5818)
(323)
B. siliquastri
dN/dS
1.00
0.75
0.50
0.25
0.00
(327)
(8)
(855)
(16)
(2087)
(47)
(5802)
(317)
C. chinensis
dN/dS
1.0
0.5
0.0
(266)
(1)
(140)
(3)
(860)
(11)
(2009)
(45)
(5493)
(264)
A X A X A X A X A X
1 2 3 4 5
(number of genes)
gene age


A
Autosomes: abdomen
X-chromosome: abdomen
Autosomes: head+thorax
X-chromosome: head+thorax
100%
80%
60%
40%
20%
0%
(302) (329) (1294) (2580) (6192)
(3) (8) (19) (57) (349)
1 2 3 4 5
B
|log2FC| of sig DE genes
(p<0.05, |log2FC|>1)
Autosomes: abdomen
Autosomes: head+thorax
X-chromosome: abdomen
X-chromosome: head+thorax
15
10
5
0
(86) (35) (82) (13) (481) (164) (711) (482) (981) (1466)
(44) (19) (43) (9) (232) (70) (324) (158) (288) (329)
(1) (1) (1) (0) (2) (4) (11) (14) (41) (61)
(1) (0) (0) (0) (1) (1) (5) (4) (15) (11)
1 2 3 4 5


A
median and standard error of dN/dS for abdominal tissue
dN/dS (max. 2)
0.250
0.225
0.200
0.175
0.150
0.125
0.100
0.075
0.050
(73)
(161)
(32)
(73)
(56)
(11)
(429)
(286)
(145)
(656)
(914)
(439)
(902)
(3218)
(1373)
1
2
3
4
5
C
pos. sel. abd.
100%
80%
60%
40%
20%
0%
(2133)
(4635)
(2000)
(54)
(195)
(75)
A
X
B
median and standard error of dN/dS for abdominal tissue
dN/dS (max. 2)
0.25
0.20
0.15
0.10
0.05
(1)
(0)
(0)
(1)
(2)
(0)
(2)
(5)
(4)
(11)
(20)
(14)
(39)
(168)
(57)
1
2
3
4
5
D
pos. sel. h+t
100%
80%
60%
40%
20%
0%
(853)
(7392)
(523)
(21)
(289)
(14)
A
X